

# Bank of Portugal Course—Empirical Macroeconomics

Jonathon Hazell  
London School of Economics

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- **Overview.** This is a course for research staff at the Bank of Portugal and economics PhD students. We will study empirical macroeconomics, with a focus on both causality and identification, both using reduced form methods and structural estimation. Lectures will often take the form of studying a topic (e.g. the effect of monetary policy on the economy) and a method (e.g. local projections and VARs).
- **Organization.** The class will meet at the following dates and times:
  - Tuesday – October 28th:
    - \* 14h00 – 17h15 (with a 15-minute break)
  - Wednesday – October 29th:
    - \* 10h00 - 12h15 (with a 15-minute break)
    - \* 14h00 – 16h15 (with a 15-minute break)
  - Thursday – October 30th:
    - \* 10h00 - 12h15 (with a 15-minute break)
    - \* 14h00 – 16h15 (with a 15-minute break)
  - Friday – October 31st:
    - \* 10h00 - 12h15 (with a 15-minute break)
    - \* 14h00 – 16h15 (with a 15-minute break)

## Rough Plan

- Tuesday Lecture: Identification in Macro, Monetary and Fiscal Policy, and Hysteresis
  - Course overview
  - Identification in macro
  - Monetary policy
  - Fiscal policy
  - Hysteresis
- Wednesday Lectures: Phillips Curves and inflation
  - Time series evidence
  - Regional estimates
  - Inflation after the Pandemic
- Thursday Lectures: labor markets

- New costs of inflation in the labor market
- Wage rigidity and unemployment fluctuations
- Friday Lectures:
  - Evidence on models of consumption behavior
    - \* Time series evidence
    - \* Studies of marginal propensity to consume
    - \* Structural estimation of precautionary savings
    - \* Introduction to event study estimation
  - Automation
    - \* Effect of robots on labor market outcomes
    - \* Introduction to shift share instruments

## References

### Useful Advice on How to do Research

I am attaching various links about how to do research. I benefitted from all of these enormously during grad school.

- **Most important.** Matthias Doepke's guide to how to give a lunch talk.
- John Cochrane's guide to writing for PhD students.
- Jesse Shapiro's advice for writing an applied micro paper and how to give an applied micro talk.
- Adam Guren's advice about how to give a lunch talk.
- Eric Zwick's 12 step guide to grad school.
- Rachael Meager's guide to public speaking.
- David Romer's 5 step guide to grad school.

### Tuesday Lecture: Identification in Macro, Monetary and Fiscal Policy, and Hysteresis

- Nakamura, Emi, and Jón Steinsson. "Identification in macroeconomics." *Journal of Economic Perspectives* 32.3 (2018): 59-86.
- Romer, C.D. and D.H. Romer (2004): "A New Measure of Monetary Shocks: Derivation and Implications," *American Economic Review*, 94(4), 1055-1084.
- Coibion, O. (2012): "Are the Effects of Monetary Policy Shocks Big or Small?" *American Economic Journal: Macroeconomics*, 4(2), 1-32.
- Ramey, Valerie A. "Macroeconomic shocks and their propagation." *Handbook of macroeconomics* 2 (2016): 71-162.
- Aruoba, S. Borağan, and Thomas Drechsel. "Identifying Monetary Policy Shocks Using Natural Language Processing." (2024).
- Nakamura, Emi, and Jón Steinsson. "High-frequency identification of monetary non-neutrality: the information effect." *The Quarterly Journal of Economics* 133.3 (2018): 1283-1330.

- Plagborg-Møller, Mikkel, and Christian K. Wolf. "Local projections and VARs estimate the same impulse responses." *Econometrica* 89.2 (2021): 955-980.
- Montiel Olea, José Luis, Plagborg-Møller, Mikkel, Qian, Eric and Wolf, Christian (2025): "Local Projections or VARs? A Primer for Macroeconomists." Working paper.
- Nakamura, E. and J. Steinsson (2014): "Fiscal Stimulus in a Monetary Union: Evidence from U.S. Regions," *American Economic Review*, 104(3), 753-792.
- Chodorow-Reich, Gabriel. "Geographic cross-sectional fiscal spending multipliers: What have we learned?." *American Economic Journal: Economic Policy* 11.2 (2019): 1-34.
- Ramey, Valerie A., and Sarah Zubairy. "Government spending multipliers in good times and in bad: evidence from US historical data." *Journal of Political Economy* 126.2 (2018): 850-901.
- Blanchard, Olivier, Eugenio Cerutti, and Lawrence Summers. "Inflation and activity—two explorations and their monetary policy implications." No. w21726. National Bureau of Economic Research, 2015.

### Wednesday Lectures: Inflation Before and After the Pandemic.

- **Required reading:** Hazell, J., Herreno, J., Nakamura, E., & Steinsson, J. (2020). The slope of the Phillips Curve: evidence from US states (No. w28005). National Bureau of Economic Research.
- Atkeson, Andrew E. Ohanian (2001). "Are Phillips Curves Useful for Forecasting Inflation?" *Federal Reserve bank of Minneapolis quarterly review* 25.1: 2-2.
- Ball, L., & Mazumder, S. (2014). A Phillips Curve with Anchored Expectations and Short-Term Unemployment. *Journal of Money, Credit and Banking*.
- Beraja, M., Hurst, E., & Ospina, J. (2019). "The aggregate implications of regional business cycles," *Econometrica*, 87(6), 1789-1833.
- Barnichon, Regis and Mesters, G. (2020). "Identifying Modern Macro Equations with Old Shocks," *The Quarterly Journal of Economics*, forthcoming.
- Gali, J. and M. Gertler (1999): "Inflation Dynamics: A Structural Econometrics Analysis," *Journal of Monetary Economics*, 44(2), 195-222.
- Mavroeidis, S., M. Plagborg-Møller, and James H. Stock (2014): "Empirical Evidence on Inflation Expectations in the New Keynesian Phillips Curve," *Journal of Economic Literature*, 52(1), 124-188.
- Beaudry, Paul, Chenyu Hou, and Franck Portier. On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation. No. w33522. National Bureau of Economic Research, 2025.
- Hazell, Jonathon, and Stephan Hobler. Do Deficits Cause Inflation?: A High Frequency Narrative Approach. London School of Economics and Political Science, 2024.

### Thursday Lectures: Labor Markets and the New Costs of Inflation.

- Guerreiro, J., Hazell, J., Lian, C., & Patterson, C. (2024). Why Do Workers Dislike Inflation?: Wage Erosion and Conflict Costs.
- Shiller, R. J. (1997). Why do people dislike inflation?. In *Reducing inflation: Motivation and strategy* (pp. 13-70). University of Chicago Press.

- Stantcheva, S. (2024). Why do we dislike inflation?. *Brookings Papers on Economic Activity*, 2004(1), 1-46.
- Pissarides, C. A. (2009). The unemployment volatility puzzle: Is wage stickiness the answer?. *Econometrica*, 77(5), 1339-1369.
- Ljungqvist, L., & Sargent, T. J. (2017). The fundamental surplus. *American Economic Review*, 107(9), 2630-65.
- Hall, Robert E. "Employment fluctuations with equilibrium wage stickiness." *American economic review* 95.1 (2005): 50-65.
- Grigsby, J., Hurst, E., & Yildirmaz, A. (2021). Aggregate nominal wage adjustments: New evidence from administrative payroll data. *American Economic Review*, 111(2), 428-71.
- Hazell, Jonathon, and Bledi Taska. "Downward rigidity in the wage for new hires." Available at SSRN 3728939 (2020).
- Gertler, Mark, Christopher Huckfeldt, and Antonella Trigari. "Unemployment fluctuations, match quality, and the wage cyclicalilty of new hires." *The Review of Economic Studies* 87.4 (2020): 1876-1914.
- Davis, Steven J., and Till Von Wachter. "Recessions and the Costs of Job Loss." *Brookings Papers on Economic Activity* 42.2 (Fall) (2011): 1-72.

#### **Friday Lecture 1: Evidence on Consumption Behavior.**

- Parker, J.A., N. Souleles, D. Johnson, and R. McClelland (2012): "Consumer Spending and the Economic Stimulus Payments of 2008," *American Economic Review*, 103(6), 2530-2553.
- Kueng, L. (2018). Excess sensitivity of high-income consumers. *The Quarterly Journal of Economics*, 133(4), 1693-1751.
- Gourinchas, P.O., and J.A. Parker (2002): "Consumption over the Life Cycle," *Econometrica*, 70(1), 47-89.
- Hsieh, C.T. (2003): "Do Consumers React to Anticipated Income Shocks? Evidence from the Alaska Permanent Fund," *American Economic Review*, 93, 397-405.
- Kueng, L. (2015): "Revisiting the Response of Household Spending to Alaska Permanent Fund Dividend using CE Data," Working Paper, Northwestern University.
- Jappelli, T., and L. Pistaferri (2010): "The Consumption Response to Income Changes," *Annual Review of Economics*, 2, 479-506.
- Johnson, D.S., J.A. Parker, and N.S. Souleles (2006): "Household Expenditure and the Income Tax Rebates of 2001," *American Economic Review*, 96, 1589-1610.
- Orchard, J., Ramey, V. A., & Wieland, J. (2022). Micro MPCs and macro counterfactuals: the case of the 2008 rebates. Working Paper.
- Borusyak, K., Jaravel, X., & Spiess, J. (2021). Revisiting event study designs: Robust and efficient estimation. *arXiv preprint arXiv:2108.12419*.

## Friday Lecture 2: Automation and Shift Share Instruments

- Acemoglu, Daron, and Pascual Restrepo. "Robots and jobs: Evidence from US labor markets." *Journal of Political Economy* 128.6 (2020): 2188-2244.
- Borusyak, K., Hull, P., & Jaravel, X. (2018). Quasi-experimental shift-share research designs (No. w24997). National Bureau of Economic Research.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. "Bartik instruments: What, when, why, and how." *American Economic Review* 110.8 (2020): 2586-2624.
- Acemoglu, D., & Autor, D. (2011). Skills, tasks and technologies: Implications for employment and earnings. In *Handbook of labor economics* (Vol. 4, pp. 1043-1171). Elsevier.
- Acemoglu, D., Autor, D., Hazell, J., & Restrepo, P. (2020). AI and jobs: Evidence from online vacancies (No. w28257). National Bureau of Economic Research.